
Task W01: Recursion Basics — Answer Sheet

Name: _____

Before You Start

This is a LaTeX source file, not a finished document; what you are reading right now is what it looks like once compiled. To edit it and produce your own PDF, you will need a LaTeX environment such as Overleaf, TeXworks, or a local TeX installation; if you need help setting it up in Linux, please let me know.

1. Keep `Task_W01_Answers.tex` and `coa-accessible-student.sty` in the same folder. This file will not compile without the style file present.
 2. Open `Task_W01_Answers.tex` in a text or code editor, not a word processor, and fill in each section below with your own work, following the instructions given in each section.
 3. Compile it by running `pdflatex Task_W01_Answers.tex`. Run this command twice in a row; the page numbers at the bottom of each page will not resolve correctly on a single pass, and will show "Page 1 of ??" instead of the real total.
 4. Every time you make a change to the `.tex` file, save it and run `pdflatex` twice again to see your latest edits in the PDF you submit.
 5. Submit both files: your edited `Task_W01_Answers.tex` source and the `Task_W01_Answers.pdf` it produces. Your diagrams are submitted separately from this file; see the reminder in each section below and the Diagram Submission Instructions in the assignment itself.
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1 Recurrence Relation Problems

Diagrams for both problems below are required, but are **not** included in this file. Submit them separately, following the Diagram Submission Instructions in the assignment. Just record your final answer here.

1.1 Problem: $B(n)$

Value of $B(6)$: _____

1.2 Problem: $D(n)$

Value of $D(6)$: _____

2 Applied Problems

Problem you are completing in full (write the number 1 through 4): _____

Final returned value for that problem: _____

The diagram for whichever problem you complete in full is required, but is **not** included in this file. Submit it separately, following the Diagram Submission Instructions in the assignment. The other three problems do not need a diagram, just your attempted solution below.

You may write pseudocode or working code in Java or C/C++ for each problem below; either is fine, and you do not need to make the same choice for every problem. If you would like to see it written the way CLRS and Walls and Mirrors write it, see the link in the assignment itself. If you use the AI-for-formatting exception described there, submit that conversation as its own file, the same way you submit your diagrams separately.

2.1 Problem 1

% Your pseudocode or code here.

2.2 Problem 2

% Your pseudocode or code here.

2.3 Problem 3

% Your pseudocode or code here.

2.4 Problem 4

% Your pseudocode or code here.

3 Bugs or Issues You Found

If you found something in the assignment that needs to be debugged, for example an unclear problem statement or an example that does not behave the way it is described, describe it here. Leave this blank if nothing came up.